The Magnetotelluric Impedance at Middelpos, South Africa

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Abstract

From 2012-07-12 through 2012-09-04 (55 days), a magnetotelluric (MT) instrument made continuous 1-second cadence measurements at a site in Middelpos, South Africa. Two methods were used to compute the surface impedance transfer function (TF): (1) 55 TFs were computed and averaged using non-overlapping one-day segments, and (2) one TF was computed using the full 55-day segment. Their error bars were calculated using parametric formulas, which assume a known distribution of errors, and a bootstrap method. In addition to coherence, another frequency–domain quality metric, the signal-to-prediction error (SE), was calculated to assess the quality of predictions of the electric field from the computed TFs. The computed TFs have significant off-diagonal components, suggesting a complex ground-conductivity structure. The SE is greater than unity only in the range of 1-minute to 6-hours for the E_x–related terms and 1-minute to 1-hour for E_y–related terms. We find differences in the transfer functions computed in the two ways that are inconsistent with the error bars. We also compare the transfer function at Middelpos with that computed at a nearby (∼34 km) location as part of a different MT campaign. We find significant differences in the transfer functions at the two locations, which suggests significant lateral variations in the ground conductivity structure near Middelpos.

Keywords: impedance tensor, magnetotelluric method, geoelectric field, geomagnetic field

1 Introduction

The electric and magnetic fields measured on Earth's surface can be related by a transfer function derived from Maxwell's equations (Vozoff (1991); Simpson and Bahr (2005); Utada et al. (2026)). The electromagnetic field measured on Earth's surface is primarily due to waves initiated in the ionosphere that penetrate into and reflect off of Earth's conducting surface at various depths (Cagniard 1953; Pirjola 1982). The transfer function at Earth's surface is referred to as the surface impedance and relates the horizontal components of the surface **E** and **B** by a 2x2 tensor with frequency dependent and complex-valued components.

The surface impedance of the Earth near conducting systems, such as power lines and pipelines, is an essential parameter for deriving geomagnetically induced currents (GICs) in the power line from measured or predicted geomagnetic fields. Given a measured or modeled geomagnetic field, the surface impedance can be used to predict the geoelectric field, which can then used to compute GICs in power lines and pipelines (e.g., Marshall et al. (2019); Alves Ribeiro et al. (2023)). The surface impedance is also the starting point for estimating the subsurface conductivity structure because the surface impedance depends on the subsurface conductivity (Simpson and Bahr 2005).

The surface impedance is typically derived from magnetotelluric (MT) measurements, which combine co-located 3-axis magnetic field measurements with 2-axis horizontal electric field measurements (Naidu (2012), Simpson and Bahr (2005)). MT measurements have been made in South Africa by various institutions. The SAMTEX campaign, comprising short-term (∼3 weeks per location) measurements along various transects in South Africa, was conducted by a consortium of geophysicists to characterize the Kaapvaal Craton in the central part of South Africa (Jones et al. 2009). The South African National Space Agency (SANSA) deployed MT measurement stations, operating for several months or more at each location, at 11 locations to provide useful data for GIC estimation by the Space Weather Centre at SANSA (Kotzé et al. 2015).

Several algorithms have been used to derive the surface impedance transfer function. Sims et al. (1971) introduced regression–based algorithms for computing the surface impedance. Variations on this method that have been used include Egbert (1997, 2011); Jones et al. (1989); Chave and Jones (2012), and Larsen et al. (1996). Recent papers that derived transfer functions from MT measurements include Jones et al. (1989), Moorkamp et al. (2007), Fujii et al. (2015), Heyns et al. (2020), Weigel and Cilliers (2019), and Chen et al. (2021).

In section 2, we provide details on the **E** and **B** time series in the time and frequency domains. In this work, we use two primary approaches to estimate the surface impedance and compare their results. In section 3, we describe these two approaches. In section 4, we present the results of the transfer function computations at Middelpos. MT measurements and data from a nearby location as part of a MT campaign in 2003 are compred with our results at Middelpos.

2 Data

Data were measured with the LEMI-417 instrument, which is designed for long–term recording of MT data; it is comprised of a 3–axis fluxgate magnetometer and a 2–axis

electric field antenna. The instrument was installed in an empty field near Middelpos, South Africa (population ∼300), located at WGS 84 geographic (latitude, longitude) (−31.906°, 20.234°) and altitude of 1165 m. The instrument is powered by batteries charged by a solar panel and is one of 11 MT instruments in southern Africa managed by SANSA (Kotzé et al. 2015). The timestamps of the measurements are from a GPS chip in the LEMI-417 instrument. Data were manually retrieved during a site visit by SANSA technicians.

The $+x$-axes of both the magnetometer and electric field probe were aligned with geomagnetic North on 2012-07-12 when the magnetic declination at Middelpos was −23.2541323°, computed using the definitive IGRF model coefficients for years 2015-2020 (Alken et al. 2021). Figure 1 is an annotated Google Earth image of the instrument location and electric field probes.



Fig. 1 Layout of the MT instrumentation at Middelpos. MID N(orth), MID S(outh), MID E(ast), and MID W(est) indicate the locations of the electric field electrodes. The length of each electrode line is 100 m. The LEMI-417 instrument is located at the crossing of the two electrode lines. The magnetometer is positioned 5 m from the LEMI-417 to reduce the impact of instrument currents on its measurements.

The time series of 1–second–cadence electric, $\mathbf{E}(t)$, and magnetic, $\mathbf{B}(t)$, field measurements is shown in Figure 2. Each component $k = x$ or y of $\mathbf{E}(t)$ was despiked by finding times when $|E_k(t+1) - E_k(t)| \geq 0.1$ mV/km and replacing values in the time range $[t-1, t+5]$ with linearly interpolated values based on $E_k(t-2)$ and $E_k(t+6)$. The number of spikes in $E_x(t)$ was 112 (0.002%); for $E_y(t)$, the number was 904 (0.02%).

Figure 3 shows the averaged amplitudes of the Fourier coefficients for each vector component of the time series shown in Figure 2. The averaged amplitudes are

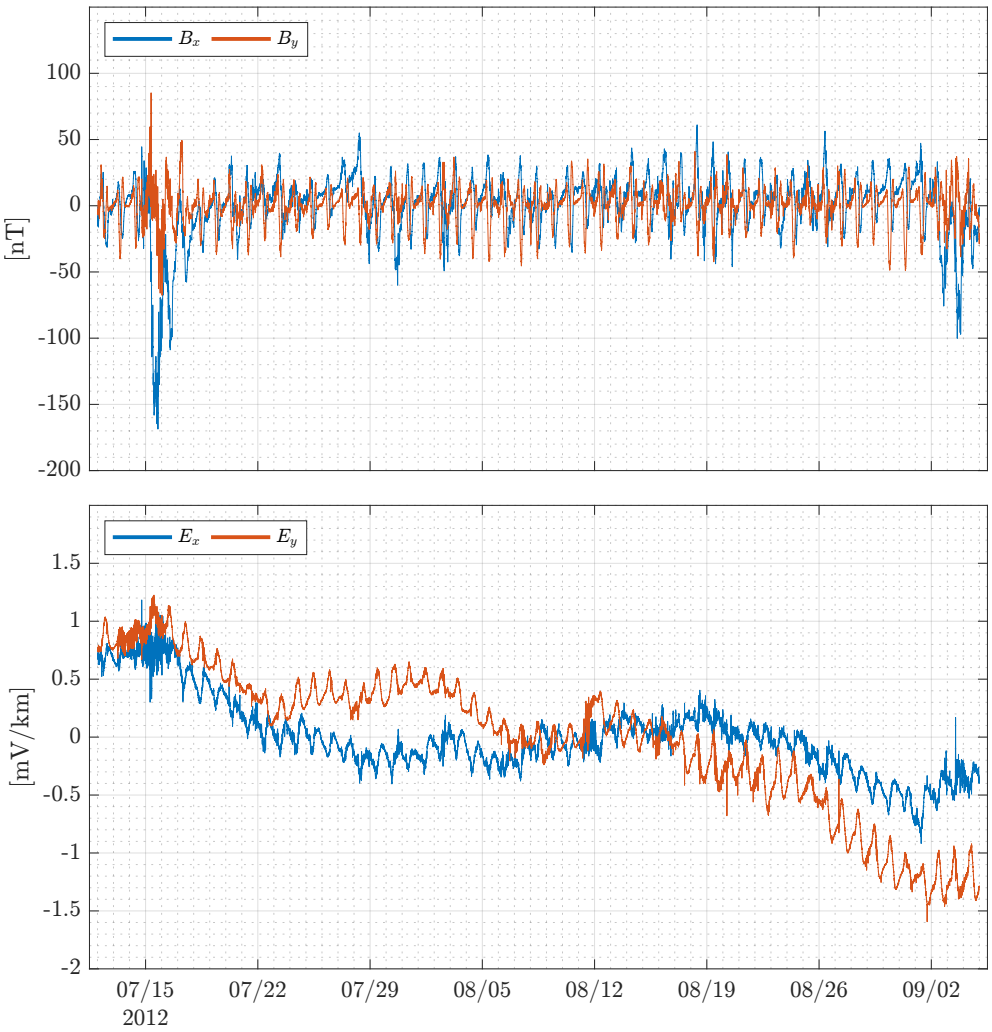


Fig. 2 Measurements at Middelpos used for analysis after mean subtraction and despiking.

frequency–domain band averages of the discrete discrete Fourier transform amplitudes. The values of the band centers, marked with filled circles, the number of values in each band, and the band boundaries are given in Table B2 of Appendix B. To compute the averages, the raw DFT coefficients were averaged within the frequency bands and the resulting f_e-centered DFT coefficients were averaged across the 55 segments. The vertical bars correspond to 95% confidence intervals computed using $\pm 1.96 s/\sqrt{55}$, where s is the sample standard deviation of the 55 f_e-centered DFT magnitudes. Error bars were also computed using a bootstrap resampling of the 55 DFT values; these error bars were within $\sim$10% of the analytical error bars.

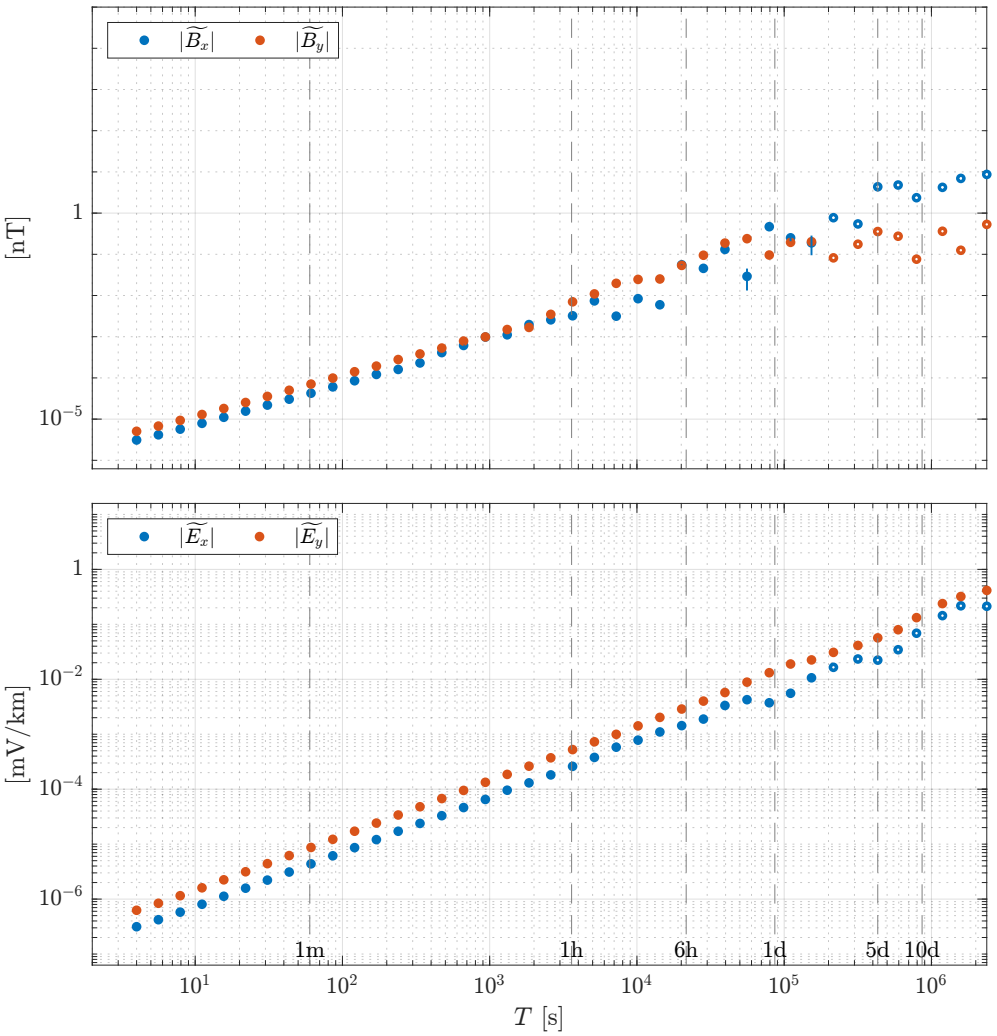


Fig. 3 Magnitudes of discrete Fourier transform coefficients for measurements in Fig 2. The vertical bars correspond to 95% confidence intervals and a filled circle indicates the computed lower bound is below zero.

3 Models and Methods

$\mathbf{E}(t)$ and $\mathbf{B}(t)$ measurements are used to estimate the complex–valued transfer function tensor $\widetilde{\boldsymbol{\mathcal{Z}}}$ in

$$\widetilde{\mathbf{E}}(f_e) = \widetilde{\boldsymbol{\mathcal{Z}}}(f_e)\widetilde{\mathbf{B}}(f_e) \ , \tag{1}$$

where f_e is an "evaluation frequency" and

$$\widetilde{\boldsymbol{\mathcal{Z}}}(f_e) = \begin{bmatrix} \widetilde{Z}_{xx}(f_e) & \widetilde{Z}_{xy}(f_e) \\ \widetilde{Z}_{yx}(f_e) & \widetilde{Z}_{yy}(f_e) \end{bmatrix} \tag{2}$$

using a method described by Sims et al. (1971): for each vector component of $\mathbf{E}$, $k = x$ or y, a four parameter over–determined linear regression is performed using

$$\widetilde{E}_k(f) = \widetilde{Z}_{kx}(f)\widetilde{B}_x(f) + \widetilde{Z}_{ky}(f)\widetilde{B}_y(f) \tag{3}$$

with values of f in a band associated with f_e. The number of parameters is four because the two regression parameters $\widetilde{Z}_{kx}$ and $\widetilde{Z}_{ky}$ in Equation 3 are complex–valued; the actual regression is performed using only real numbers (Egbert and Booker 1986). The values of f are those associated with Fourier coefficients of $E_k(t)$ and $B_k(t)$, $\widetilde{E}_k(f)$ and $\widetilde{B}_k(f)$. This regression method corresponds to minimizing the least-squared error between the predicted and measured $\widetilde{E}_k(f)$ values in each band.

Historically, two methods have been used for computing the Fourier coefficients for uniform cadence time series – one uses the standard DFT (Discrete Fourier Transform) formula for equally spaced records, typically implemented using the FFT algorithm. The other is an approximation based on the method of "Cascade Decimation" (Wight and Bostick 1980), which can require less memory and produces logarithmically spaced evaluation frequencies. Here, we use the standard DFT formula because it is not an approximation and memory is not a constraint for the data considered.

There are two general methods for computing estimates of $\widetilde{Z}_{kx}$ and $\widetilde{Z}_{ky}$ for each evaluation frequency when there are not significant data gaps: (1) compute the Fourier coefficients using the full time series and then perform regression using the Fourier coefficients in a band centered on the evaluation frequency and (2) split the full time series into (possibly overlapping) segments, perform a regression using the data in each segment to find segment estimates of $\widetilde{Z}_{kx}(f_e)$ and $\widetilde{Z}_{ky}(f_e)$, and then average the segment estimates of $\widetilde{Z}_{kx}(f_e)$ and $\widetilde{Z}_{ky}(f_e)$. In this paper, we present results from both methods; for the second method, non–overlapping time series segments were used, and the estimates of $\widetilde{Z}_{kx}(f_e)$ and $\widetilde{Z}_{ky}(f_e)$ are unweighted averages.

A key result is that there are significant differences in the transfer functions computed using these two approaches. We have verified that the two approaches should give transfer functions without statistically significant differences by generating a simulated $\mathbf{E}(t)$ by convolving gaussian distributed white noise $\mathbf{B}(t)$ with a known transfer function $\widetilde{\boldsymbol{\mathcal{Z}}}$ N times. The two approaches will not give identical transfer functions due to the differences in frequency grids, but the averaged and interpolated transfer functions for large N converges. As a result, we conclude that the observed differences are likely due to non-stationarity.

In the literature, results are often presented using only one algorithm and a single set of parameters for that algorithm. However, it seems that, given that these two reasonable approaches for computing a transfer function can yield different results, there is an "algorithmic uncertainty" that should be accounted for by computing the transfer function using different algorithms and varying parameter values for a given algorithm. There is also an uncertainty due to the data selection; Chen et al. (2022) noted that for some SAMTEX sites, the transfer functions were dependent on

the time intervals used in their computation, in particular on whether the interval contained a period of high geomagnetic activity. This is expected because the plane wave approximation is invalid when geomagnetic activity is strong. Conversely, in the limit of low geomagnetic activity, the signal-to-noise ratio is low, leading to uncertainty in the computed transfer functions.

There are two common modifications to these estimation methods. The first is the "remote reference" method, which uses $\mathbf{B}(t)$ measurements covering the same time interval from a nearby magnetometer (Simpson and Bahr 2005). In our case, we do not have such measurements. The second uses a robust algorithm for regression, which involves computing estimates of $\widetilde{Z}_{kx}$ and $\widetilde{Z}_{ky}$ using a custom error function instead of the standard sum–of–squares error function (Simpson and Bahr (2005); see also Thomson and Vernon (2016) for a discussion about the "overselling" of this method). The custom error function automatically down-weights outliers. For the data considered, we have found that this method produces differing results only at evaluation frequencies where the signal-to-prediction error, described in Section 4 is less than 1.0.

4 Results

Figure 4 shows the results of the regression described in Section 3. For the components of $\widetilde{\boldsymbol{\mathcal{Z}}}$ computed using one 55–day segment, 95% confidence intervals were computed using formulas that assume the regression errors are independent and identically distributed (Draper and Smith (1981); chapter 2.6).

For the magnitudes calculated using the average of 55 one–day segments, the 95% confidence intervals were calculated using propagation of errors on $|\widetilde{Z}| = \sqrt{Z_{\mathrm{r}}^2 + Z_{\mathrm{i}}^2}$,

$$\Delta\left|\widetilde{Z}\right| = \sqrt{(Z_{\mathrm{r}}\Delta Z_{\mathrm{r}})^2 + (Z_{\mathrm{i}}\Delta Z_{\mathrm{i}})^2}/\left|\widetilde{Z}\right| \tag{4}$$

where ΔZ_{r} and ΔZ_{i} are the lengths of the 95% confidence intervals for the real, Z_{r}, and imaginary, Z_{i}, parts of $\widetilde{Z}$, respectively; $\widetilde{Z}$ is one of the $\widetilde{B}_x$, $\widetilde{B}_y$, $\widetilde{E}_x$, and $\widetilde{E}_y$, which are frequency domain averages shown in Figure 3. The lengths of the symmetric 95% confidence intervals for the averages Z_{r} and Z_{i} were determined using $\pm 2(s/\sqrt{n})t_{n-1}^{-1}(0.025)$, where n is the number of values used to calculate the average, $t_{n-1}^{-1}(0.025)$ is the inverse t-distribution evaluated at 0.025, and s is the standard deviation of the sample.

The error bars for the phases were calculated using propagation of errors on

$$\phi_{ij} = \tan^{-1}[\mathrm{Im}(\widetilde{Z}_{ij})/\mathrm{Re}(\widetilde{Z}_{ij})],$$

which gives

$$\Delta\phi_{ij} = (1/|\widetilde{Z}_{ij}|^2)\sqrt{\left[\mathrm{Im}(\widetilde{Z}_{ij})\mathrm{Re}(\Delta\widetilde{Z}_{ij})\right]^2 + \left[\mathrm{Re}(\widetilde{Z}_{ij})\mathrm{Im}(\Delta\widetilde{Z}_{ij})\right]^2},$$

where $\Delta\widetilde{Z}_{ij}$ is the 95% confidence interval length for $\widetilde{Z}_{ij}$, $|\cdot|$ is the complex magnitude, and $\mathrm{Re}(\cdot)$ and $\mathrm{Im}(\cdot)$ are the real and imaginary parts of the argument, respectively.

The quality of the computed transfer functions were assessed using the traditional coherence metric and also a signal–to–error metric.

The squared coherence, $\mathrm{Coh}^2_{XY}(f_e)$, was computed using

$$\mathrm{Coh}^2_{XY}(f_e) = \frac{\left|\sum_{j=1}^{N_e} \widetilde{X}^*(f_j)\widetilde{Y}(f_j)\right|^2}{\sum_{j=1}^{N_e}\left|\widetilde{X}(f_j)\right|^2 \sum_{j=1}^{N_e}\left|\widetilde{Y}(f_j)\right|^2}, \tag{5}$$

where j is the index of the DFT frequency in the band centered on $f_e = 1/T_e$; Tables B2 and B1 in Appendix B list the band centers, ranges, and number of frequencies, N_e, in each band for each regression method.

95% confidence intervals were computed using bootstrap resampling: for each f_e band, the N_e pairs of $\widetilde{Z}_{kx}$ and $\widetilde{Z}_{ky}$ were randomly selected with replacement, and the sums in the Coh^2 formula were computed using these randomly selected pairs to compute a bootstrap coherence Coh^{2*}. This process was repeated $1,000$ times, and the results were sorted in ascending order; the maximum SE error bar value is the 975th value in the sorted Coh^{2*} list, and the minimum is the 25th. No confidence intervals were computed if $N_e \leq 10$.

Traditionally, the primary quality metric used for evaluating the transfer function is $\mathrm{Coh}^2_{E_x B_y}$ and $\mathrm{Coh}^2_{E_y B_x}$. A limitation of this metric is that it does not directly quantify how well the Fourier coefficients of the predicted electric field, $\widetilde{E}^p_k$, match those of the measured electric field, $\widetilde{E}_k$. It quantifies only the coherence between the measured quantities. As a result, we have also computed the coherence between the Fourier coefficients of the measured and predicted electric field, $\mathrm{Coh}^2_{E_k E^{\mathrm{p}}_k}$ ($k = x$ or y), as it is a more direct measure of the ability of the transfer function to reproduce the signal.

Although $\mathrm{Coh}^2_{E_k E^{\mathrm{p}}_k}$ is a more direct measure of how well a transfer function reproduces the measured electric field is still has a limitation — two signals that are identical except for a scaling factor will have the same coherence for any scaling factor. As a result, we have introduced the signal–to–error, which can be used to make a more direct assessment of how well the estimated transfer function captures the measured signal.

The signal–to–error ratio, $\mathrm{SE}_k(f_e)$, for component $k = x$ or y of $\mathbf{E}$ was calculated using

$$\mathrm{SE}_k(f_e) = \frac{\sum_{j=1}^{N_e}\left|\widetilde{E}_k(f_j)\right|^2}{\sum_{j=1}^{N_e}\left|\Delta\widetilde{E}_k(f_j)\right|^2}, \tag{6}$$

where the summation frequencies and f_e values are the same as those used for computing Coh.

The error, $\Delta\widetilde{E}_k = \widetilde{E}^{\mathrm{p}}_k - \widetilde{E}_k$, is the difference between the complex–valued DFT amplitudes of the predicted, $E^{\mathrm{p}}_k(t)$, and measured, $E_k(t)$, signals after convolution

with a Parzen window. For both regression methods, the predicted signal, E_k^{p}, was computed by linearly interpolating the real and imaginary parts of $\widetilde{Z}_{kx}$ and $\widetilde{Z}_{ky}$ onto a uniform frequency grid of $N = 86400 \times 55$ points. At interpolation grid frequencies outside the range of f_e values, the interpolated values of $\widetilde{Z}_{kx}$ and $\widetilde{Z}_{ky}$ were set to zero. 95% error bars were computed using the same as method used for Coh.

Note that SE defined by Equation 6 differs in general from the signal–to–noise ratio, $\mathrm{Coh}^2/(1 - \mathrm{Coh}^2)$ that applies only if the input signal and the noise in the output signal (which is unknown) are uncorrelated and the input signal has no noise (Miranda de Sá (2006); Pedersen (1988)).

The squared coherence and signal-to-error associated with the transfer functions shown in Figure 4 are shown in Figure 5. In the top panel of Figure 5(a), we see that the transfer function for E_x only captures the variations in the data at periods longer than approximately 40 seconds, and that the model computed using one 55-day segment has a slightly higher SE ratio. For E_y, the top panel of Figure 5(b) shows that the period range over which the transfer function produces useful predictions is from approximately 6 minutes to 6 hours.

5 Comparison to KAP103

Jones et al. (2009) presented a summary of an MT survey of sites in South Africa; the results of the transfer function calculations are available from Jones (2004), which were computed using the Bounded Influence Remote Reference algorithm (Chave and Thomson 2004). The site nearest Middelpos is KAP103, which is shown in Figure 6.

In Figure 7, the 26–day magnetic and electric time series are shown along; the magnitude of their Fourier coefficients is shown in Figure 8. The measurement interval included the large geomagnetic storm of November 20–22, 2003, that followed the October 31st "Halloween" storm (Gopalswamy et al. 2005).

In Figure 9, we compare the pre–computed KAP103 transfer function (Jones (2004)) with that computed at Middelpos (corresponding to the "1 55–day segment" lines shown in Figure 4). In addition, we have used the time series shown in Figure 7 and the single transfer methodology described in the previous section.

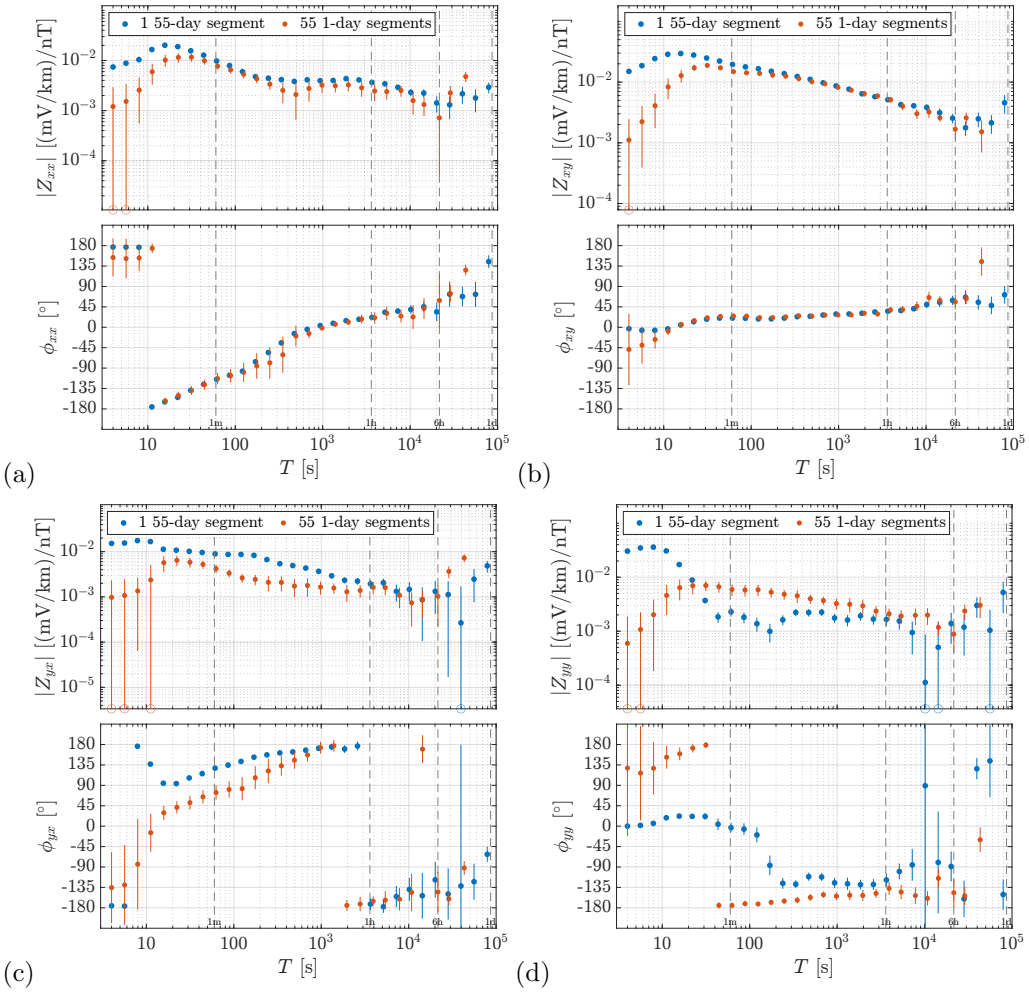


Fig. 4 Magnitudes and phases for components of $\widetilde{\boldsymbol{Z}}$: (a) $|\widetilde{Z}_{xx}|$ and ϕ_{xx}; (b) $|\widetilde{Z}_{xy}|$ and ϕ_{xy}; (c) $|\widetilde{Z}_{yx}|$ and ϕ_{yx}; and (d) $|\widetilde{Z}_{yy}|$ and ϕ_{yy}. Open circles on the lower part of the impedance amplitude error bars indicate that this formula yielded a value below zero.

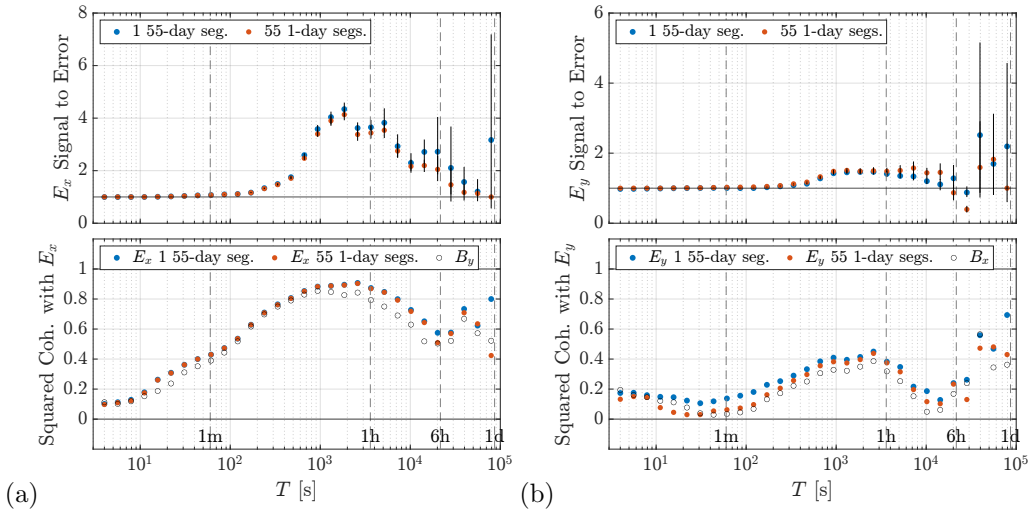


Fig. 5 Model metrics for (a) E_x, which depends on $\widetilde{Z}_{xx}$ and $\widetilde{Z}_{xy}$ shown in Figure 4(a)-(b) and (b) E_y, which depends on $\widetilde{Z}_{yx}$ and $\widetilde{Z}_{yy}$ shown in Figure 4(b)-(c).

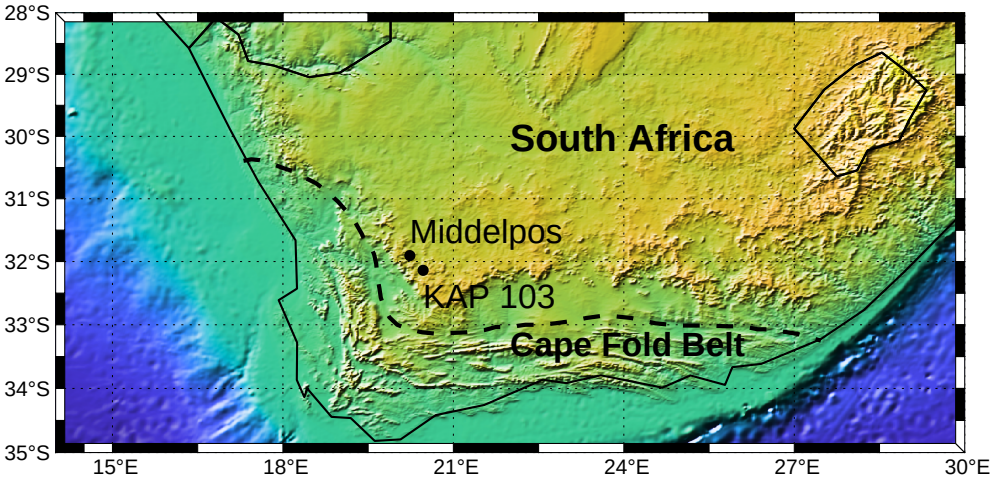


Fig. 6 Locations of Middelpos (Geographic [latitude, longitude] = [−31.906°, 20.234°]) and KAP103 ([−32.139°, 20.468°]) MT station. The relief map is from the ETOPO1 Global Relief Model (NOAA/NCEI (2006)), and the Cape Fold Belt line was derived from the digitization of Figure 1 of Nguuri et al. (2001)

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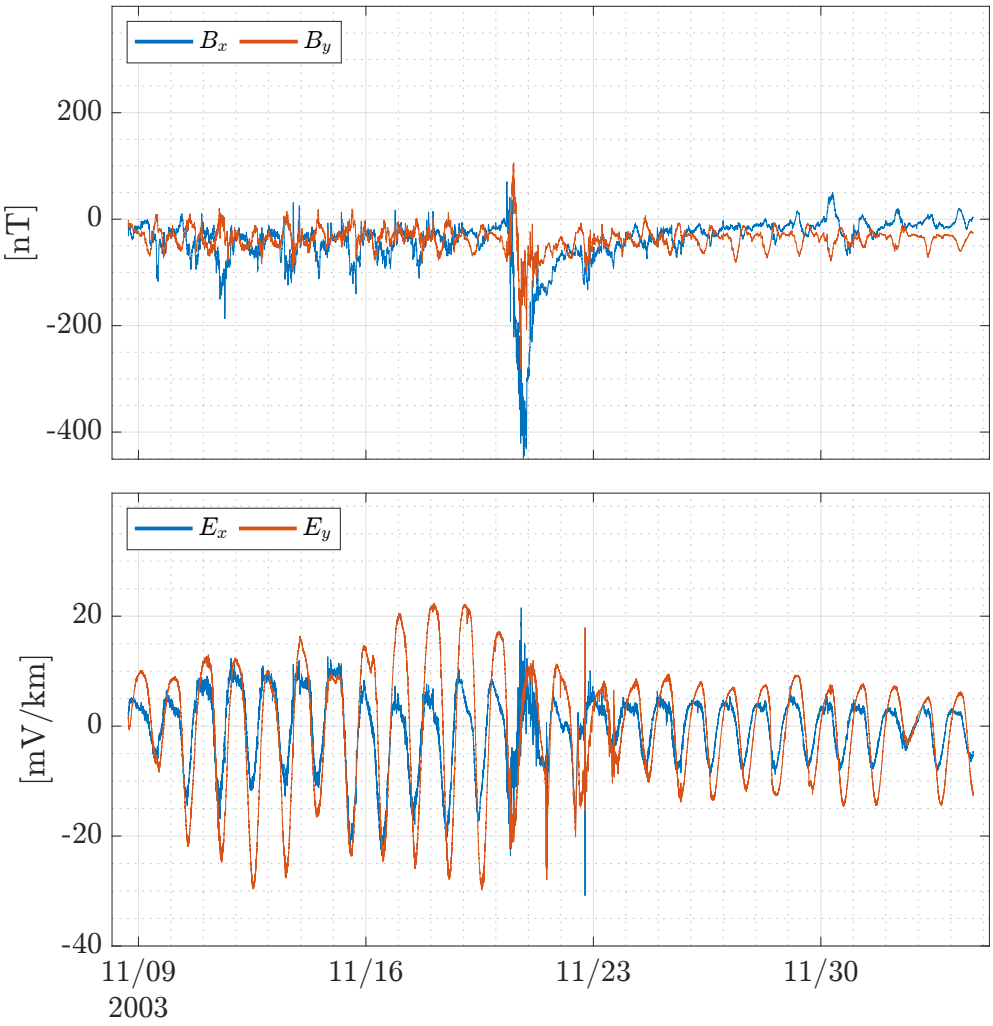


Fig. 7 Measurements from KAP-103 used for analysis after mean subtraction and despiking.

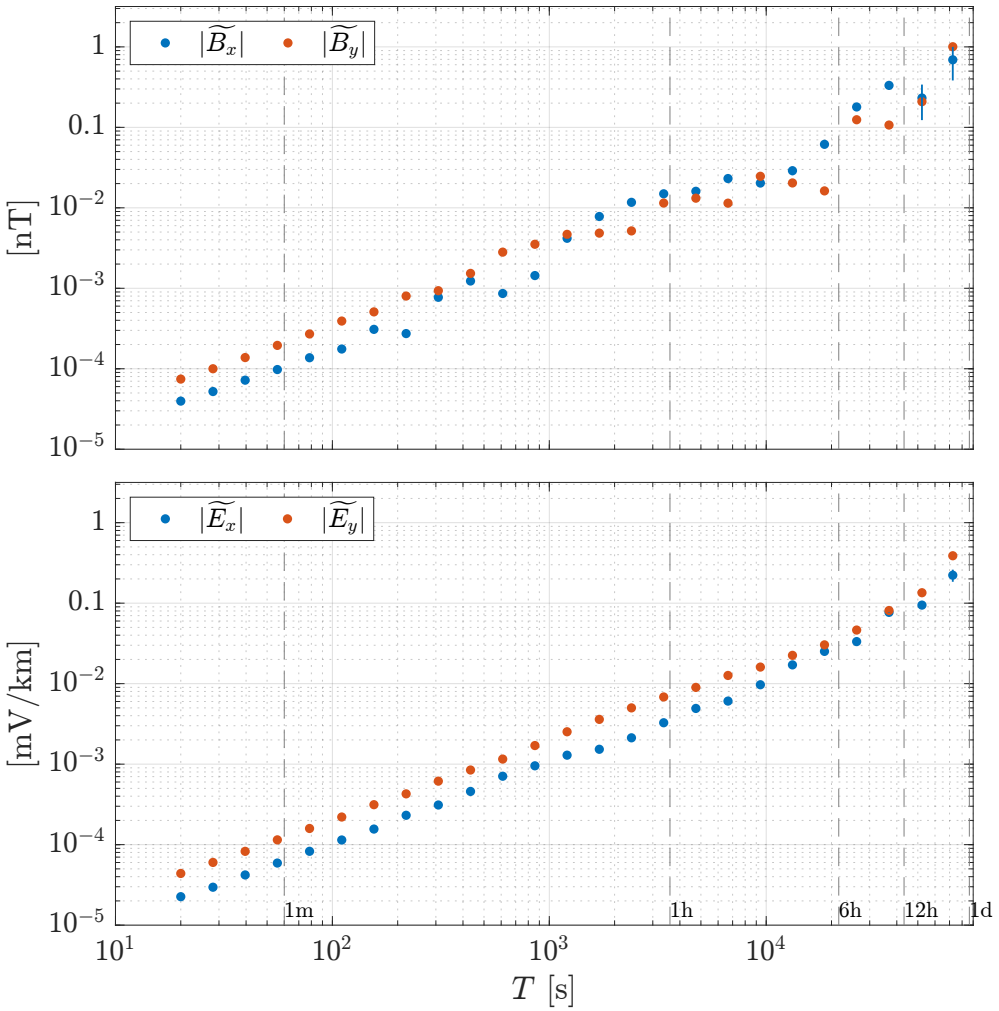


Fig. 8 Magnitudes of discrete Fourier transform coefficients for measurements in 7. The vertical bars correspond to 95% confidence intervals.

There are several significant differences in the transfer functions shown in Figure 9. First, as in Figure 4, the method used to compute the transfer function can lead to statistically significant differences in the transfer functions (with respect to the computed error bars). In the case of the comparison of the BIRRP transfer functions with the single transfer function method used in the previous section, significant differences are found in all components of $\widetilde{\boldsymbol{\mathcal{Z}}}$, and these differences tend to occur at short and long periods.

Much larger differences exist between the amplitudes of the transfer functions computed at Middelpos and at KAP103, where the magnitudes of the Middelpos transfer function components range from 2 to 10 times smaller than those at KAP103.

However, the transfer function phases are generally consistent, with the exception of $\widetilde{Z}_{yy}$.

To determine if the large geomagnetic storm may explain some of the differences, we have computed transfer functions using only data prior to November 20th and find consistent results, similar to Chen et al. (2022), who found that for several of the SAMTEX sites, the computed transfer functions were not strongly dependent on whether the storm was included or not.

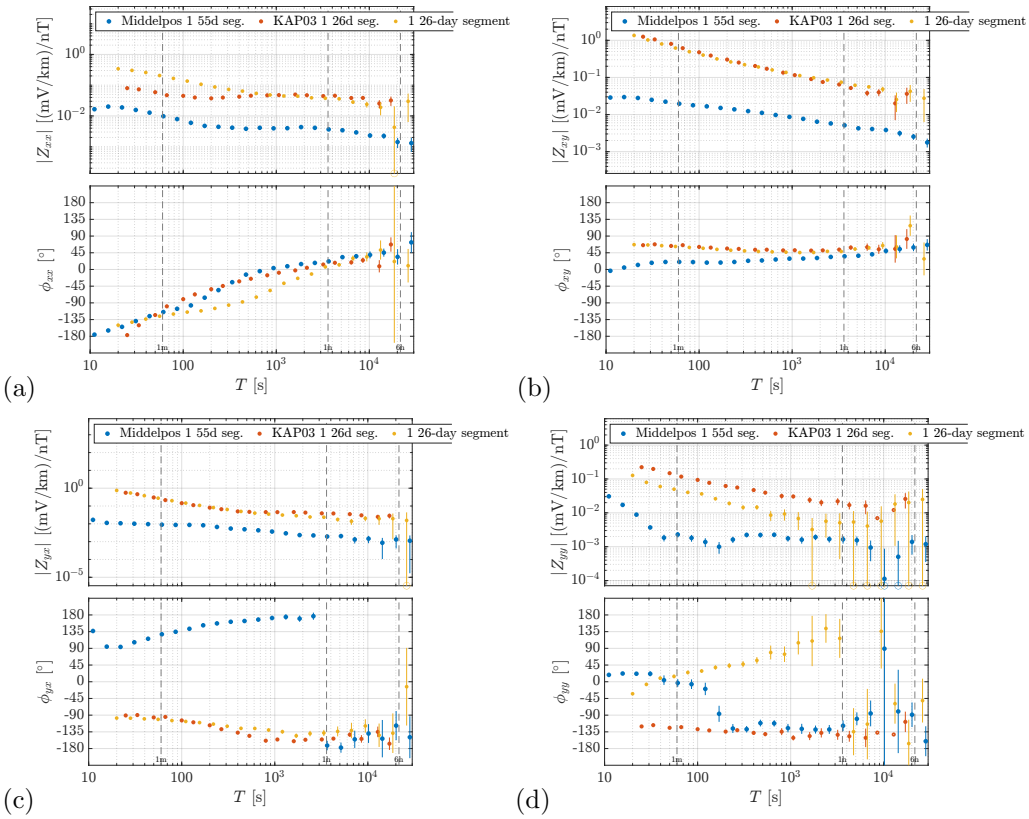


Fig. 9 Comparison of transfer function computed at Middelpos with two transfer functions computed at the KAP103 site. Error bars have the format as in Figure 4.

6 Summary

Transfer functions (surface impedances) were computed from magnetotelluric data collected at Middelpos, South Africa, over 55 days. We find that the computed transfer function depends on the method used to perform the regression. In one case, the full time range of data was split into 55 1-day segments, and the resulting 55 transfer functions were averaged in the frequency domain. In the other case, no splitting was performed, and a single transfer function was computed. At many frequencies, and for all components of the transfer function, differences were found that were inconsistent with the computed error bars.

We also compared the transfer function at Middelpos with the nearest MT site from the SAMTEX campaign. The transfer function magnitudes differed significantly at all frequencies.

Appendix A The Middelpos MT Instrument

(a)  (b) 

Fig. A1
The Middelpos MT instrument. (a) The LEMI-417 instrument and the 12 V lead-acid batteries that power it are in a subterranean enclosure with concrete walls. The instrument is mounted on two 25-liter water bottles to reduce temperature variation. (b) The solar panels are mounted on a steel frame over the enclosure. The side openings are covered with a canvas to prevent direct sunlight on the instrument. The small white dome above the solar panels is the GPS antenna.

Appendix B Evaluation Frequencies

Band center frequencies (and periods), f_e (T_e), lower and upper band bounds, f_l and f_u (T_l and T_u), lower and upper band DFT indices, i_l and i_u, and number of frequencies in band, N_e for the 1-day and 55-day regression methods are shown in Tables B1 and B2, respectively.

Table B1 Frequency domain band information for the 1-day regression method.

f_e [Hz]	f_l [Hz]	f_u [Hz]	T_e [s]	T_l [s]	T_u [s]	i_l	i_u	N_e
$2.31{\cdot}10^{-5}$	$1.16{\cdot}10^{-5}$	$3.47{\cdot}10^{-5}$	$4.32{\cdot}10^{4}$	$2.88{\cdot}10^{4}$	$8.64{\cdot}10^{4}$	2	4	3
$3.47{\cdot}10^{-5}$	$2.31{\cdot}10^{-5}$	$4.63{\cdot}10^{-5}$	$2.88{\cdot}10^{4}$	$2.16{\cdot}10^{4}$	$4.32{\cdot}10^{4}$	3	5	3
$4.63{\cdot}10^{-5}$	$3.47{\cdot}10^{-5}$	$5.79{\cdot}10^{-5}$	$2.16{\cdot}10^{4}$	$1.73{\cdot}10^{4}$	$2.88{\cdot}10^{4}$	4	6	3
$6.94{\cdot}10^{-5}$	$4.63{\cdot}10^{-5}$	$9.26{\cdot}10^{-5}$	$1.44{\cdot}10^{4}$	$1.08{\cdot}10^{4}$	$2.16{\cdot}10^{4}$	5	9	5
$9.26{\cdot}10^{-5}$	$6.94{\cdot}10^{-5}$	$1.16{\cdot}10^{-4}$	$1.08{\cdot}10^{4}$	$8.64{\cdot}10^{3}$	$1.44{\cdot}10^{4}$	7	11	5
$1.27{\cdot}10^{-4}$	$9.26{\cdot}10^{-5}$	$1.62{\cdot}10^{-4}$	$7.85{\cdot}10^{3}$	$6.17{\cdot}10^{3}$	$1.08{\cdot}10^{4}$	9	15	7
$1.85{\cdot}10^{-4}$	$1.27{\cdot}10^{-4}$	$2.43{\cdot}10^{-4}$	$5.40{\cdot}10^{3}$	$4.11{\cdot}10^{3}$	$7.85{\cdot}10^{3}$	12	22	11
$2.55{\cdot}10^{-4}$	$1.85{\cdot}10^{-4}$	$3.24{\cdot}10^{-4}$	$3.93{\cdot}10^{3}$	$3.09{\cdot}10^{3}$	$5.40{\cdot}10^{3}$	17	29	13
$3.59{\cdot}10^{-4}$	$2.55{\cdot}10^{-4}$	$4.63{\cdot}10^{-4}$	$2.79{\cdot}10^{3}$	$2.16{\cdot}10^{3}$	$3.93{\cdot}10^{3}$	23	41	19
$5.09{\cdot}10^{-4}$	$3.59{\cdot}10^{-4}$	$6.60{\cdot}10^{-4}$	$1.96{\cdot}10^{3}$	$1.52{\cdot}10^{3}$	$2.79{\cdot}10^{3}$	32	58	27
$7.18{\cdot}10^{-4}$	$5.09{\cdot}10^{-4}$	$9.26{\cdot}10^{-4}$	$1.39{\cdot}10^{3}$	$1.08{\cdot}10^{3}$	$1.96{\cdot}10^{3}$	45	81	37
$1.02{\cdot}10^{-3}$	$7.18{\cdot}10^{-4}$	$1.32{\cdot}10^{-3}$	$9.82{\cdot}10^{2}$	$7.58{\cdot}10^{2}$	$1.39{\cdot}10^{3}$	63	115	53
$1.44{\cdot}10^{-3}$	$1.02{\cdot}10^{-3}$	$1.85{\cdot}10^{-3}$	$6.97{\cdot}10^{2}$	$5.40{\cdot}10^{2}$	$9.82{\cdot}10^{2}$	89	161	73
$2.03{\cdot}10^{-3}$	$1.44{\cdot}10^{-3}$	$2.62{\cdot}10^{-3}$	$4.94{\cdot}10^{2}$	$3.82{\cdot}10^{2}$	$6.97{\cdot}10^{2}$	125	227	103
$2.86{\cdot}10^{-3}$	$2.03{\cdot}10^{-3}$	$3.69{\cdot}10^{-3}$	$3.50{\cdot}10^{2}$	$2.71{\cdot}10^{2}$	$4.94{\cdot}10^{2}$	176	320	145
$4.03{\cdot}10^{-3}$	$2.86{\cdot}10^{-3}$	$5.20{\cdot}10^{-3}$	$2.48{\cdot}10^{2}$	$1.92{\cdot}10^{2}$	$3.50{\cdot}10^{2}$	248	450	203
$5.68{\cdot}10^{-3}$	$4.03{\cdot}10^{-3}$	$7.34{\cdot}10^{-3}$	$1.76{\cdot}10^{2}$	$1.36{\cdot}10^{2}$	$2.48{\cdot}10^{2}$	349	635	287
$8.02{\cdot}10^{-3}$	$5.68{\cdot}10^{-3}$	$1.04{\cdot}10^{-2}$	$1.25{\cdot}10^{2}$	$9.65{\cdot}10^{1}$	$1.76{\cdot}10^{2}$	492	896	405
$1.13{\cdot}10^{-2}$	$8.02{\cdot}10^{-3}$	$1.46{\cdot}10^{-2}$	$8.84{\cdot}10^{1}$	$6.85{\cdot}10^{1}$	$1.25{\cdot}10^{2}$	694	1262	569
$1.59{\cdot}10^{-2}$	$1.13{\cdot}10^{-2}$	$2.06{\cdot}10^{-2}$	$6.27{\cdot}10^{1}$	$4.86{\cdot}10^{1}$	$8.84{\cdot}10^{1}$	978	1780	803
$2.25{\cdot}10^{-2}$	$1.59{\cdot}10^{-2}$	$2.91{\cdot}10^{-2}$	$4.44{\cdot}10^{1}$	$3.44{\cdot}10^{1}$	$6.27{\cdot}10^{1}$	1379	2511	1133
$3.17{\cdot}10^{-2}$	$2.25{\cdot}10^{-2}$	$4.10{\cdot}10^{-2}$	$3.15{\cdot}10^{1}$	$2.44{\cdot}10^{1}$	$4.44{\cdot}10^{1}$	1945	3541	1597
$4.48{\cdot}10^{-2}$	$3.17{\cdot}10^{-2}$	$5.78{\cdot}10^{-2}$	$2.23{\cdot}10^{1}$	$1.73{\cdot}10^{1}$	$3.15{\cdot}10^{1}$	2743	4995	2253
$6.31{\cdot}10^{-2}$	$4.48{\cdot}10^{-2}$	$8.15{\cdot}10^{-2}$	$1.58{\cdot}10^{1}$	$1.23{\cdot}10^{1}$	$2.23{\cdot}10^{1}$	3869	7045	3177
$8.91{\cdot}10^{-2}$	$6.31{\cdot}10^{-2}$	$1.15{\cdot}10^{-1}$	$1.12{\cdot}10^{1}$	$8.69{\cdot}10^{0}$	$1.58{\cdot}10^{1}$	5457	9939	4483
$1.26{\cdot}10^{-1}$	$8.91{\cdot}10^{-2}$	$1.62{\cdot}10^{-1}$	$7.96{\cdot}10^{0}$	$6.16{\cdot}10^{0}$	$1.12{\cdot}10^{1}$	7698	14016	6319
$1.77{\cdot}10^{-1}$	$1.26{\cdot}10^{-1}$	$2.29{\cdot}10^{-1}$	$5.64{\cdot}10^{0}$	$4.37{\cdot}10^{0}$	$7.96{\cdot}10^{0}$	10857	19771	8915
$2.50{\cdot}10^{-1}$	$1.77{\cdot}10^{-1}$	$3.23{\cdot}10^{-1}$	$4.00{\cdot}10^{0}$	$3.10{\cdot}10^{0}$	$5.64{\cdot}10^{0}$	15314	27888	12575

Table B2 Frequency domain band information for the 55-day regression method.

f_e [Hz]	f_l [Hz]	f_u [Hz]	T_e [s]	T_l [s]	T_u [s]	i_l	i_u	N_e
$4.21{\cdot}10^{-7}$	$2.10{\cdot}10^{-7}$	$6.31{\cdot}10^{-7}$	$2.38{\cdot}10^{6}$	$1.58{\cdot}10^{6}$	$4.75{\cdot}10^{6}$	2	4	3
$6.31{\cdot}10^{-7}$	$4.21{\cdot}10^{-7}$	$8.42{\cdot}10^{-7}$	$1.58{\cdot}10^{6}$	$1.19{\cdot}10^{6}$	$2.38{\cdot}10^{6}$	3	5	3
$8.42{\cdot}10^{-7}$	$6.31{\cdot}10^{-7}$	$1.05{\cdot}10^{-6}$	$1.19{\cdot}10^{6}$	$9.50{\cdot}10^{5}$	$1.58{\cdot}10^{6}$	4	6	3
$1.26{\cdot}10^{-6}$	$8.42{\cdot}10^{-7}$	$1.68{\cdot}10^{-6}$	$7.92{\cdot}10^{5}$	$5.94{\cdot}10^{5}$	$1.19{\cdot}10^{6}$	5	9	5
$1.68{\cdot}10^{-6}$	$1.26{\cdot}10^{-6}$	$2.10{\cdot}10^{-6}$	$5.94{\cdot}10^{5}$	$4.75{\cdot}10^{5}$	$7.92{\cdot}10^{5}$	7	11	5
$2.31{\cdot}10^{-6}$	$1.68{\cdot}10^{-6}$	$2.95{\cdot}10^{-6}$	$4.32{\cdot}10^{5}$	$3.39{\cdot}10^{5}$	$5.94{\cdot}10^{5}$	9	15	7
$3.16{\cdot}10^{-6}$	$2.31{\cdot}10^{-6}$	$4.00{\cdot}10^{-6}$	$3.17{\cdot}10^{5}$	$2.50{\cdot}10^{5}$	$4.32{\cdot}10^{5}$	12	20	9
$4.63{\cdot}10^{-6}$	$3.16{\cdot}10^{-6}$	$6.10{\cdot}10^{-6}$	$2.16{\cdot}10^{5}$	$1.64{\cdot}10^{5}$	$3.17{\cdot}10^{5}$	16	30	15
$6.52{\cdot}10^{-6}$	$4.63{\cdot}10^{-6}$	$8.42{\cdot}10^{-6}$	$1.53{\cdot}10^{5}$	$1.19{\cdot}10^{5}$	$2.16{\cdot}10^{5}$	23	41	19
$9.05{\cdot}10^{-6}$	$6.52{\cdot}10^{-6}$	$1.16{\cdot}10^{-5}$	$1.11{\cdot}10^{5}$	$8.64{\cdot}10^{4}$	$1.53{\cdot}10^{5}$	32	56	25
$1.26{\cdot}10^{-5}$	$9.05{\cdot}10^{-6}$	$1.62{\cdot}10^{-5}$	$7.92{\cdot}10^{4}$	$6.17{\cdot}10^{4}$	$1.11{\cdot}10^{5}$	44	78	35
$1.79{\cdot}10^{-5}$	$1.26{\cdot}10^{-5}$	$2.31{\cdot}10^{-5}$	$5.59{\cdot}10^{4}$	$4.32{\cdot}10^{4}$	$7.92{\cdot}10^{4}$	61	111	51
$2.53{\cdot}10^{-5}$	$1.79{\cdot}10^{-5}$	$3.26{\cdot}10^{-5}$	$3.96{\cdot}10^{4}$	$3.07{\cdot}10^{4}$	$5.59{\cdot}10^{4}$	86	156	71
$3.54{\cdot}10^{-5}$	$2.53{\cdot}10^{-5}$	$4.55{\cdot}10^{-5}$	$2.83{\cdot}10^{4}$	$2.20{\cdot}10^{4}$	$3.96{\cdot}10^{4}$	121	217	97
$4.97{\cdot}10^{-5}$	$3.54{\cdot}10^{-5}$	$6.40{\cdot}10^{-5}$	$2.01{\cdot}10^{4}$	$1.56{\cdot}10^{4}$	$2.83{\cdot}10^{4}$	169	305	137
$6.99{\cdot}10^{-5}$	$4.97{\cdot}10^{-5}$	$9.01{\cdot}10^{-5}$	$1.43{\cdot}10^{4}$	$1.11{\cdot}10^{4}$	$2.01{\cdot}10^{4}$	237	429	193
$9.83{\cdot}10^{-5}$	$6.99{\cdot}10^{-5}$	$1.27{\cdot}10^{-4}$	$1.02{\cdot}10^{4}$	$7.89{\cdot}10^{3}$	$1.43{\cdot}10^{4}$	333	603	271
$1.38{\cdot}10^{-4}$	$9.83{\cdot}10^{-5}$	$1.78{\cdot}10^{-4}$	$7.23{\cdot}10^{3}$	$5.61{\cdot}10^{3}$	$1.02{\cdot}10^{4}$	468	848	381
$1.94{\cdot}10^{-4}$	$1.38{\cdot}10^{-4}$	$2.51{\cdot}10^{-4}$	$5.14{\cdot}10^{3}$	$3.99{\cdot}10^{3}$	$7.23{\cdot}10^{3}$	658	1192	535
$2.74{\cdot}10^{-4}$	$1.94{\cdot}10^{-4}$	$3.53{\cdot}10^{-4}$	$3.66{\cdot}10^{3}$	$2.84{\cdot}10^{3}$	$5.14{\cdot}10^{3}$	925	1677	753
$3.85{\cdot}10^{-4}$	$2.74{\cdot}10^{-4}$	$4.96{\cdot}10^{-4}$	$2.60{\cdot}10^{3}$	$2.02{\cdot}10^{3}$	$3.66{\cdot}10^{3}$	1301	2357	1057
$5.41{\cdot}10^{-4}$	$3.85{\cdot}10^{-4}$	$6.97{\cdot}10^{-4}$	$1.85{\cdot}10^{3}$	$1.43{\cdot}10^{3}$	$2.60{\cdot}10^{3}$	1829	3313	1485
$7.61{\cdot}10^{-4}$	$5.41{\cdot}10^{-4}$	$9.80{\cdot}10^{-4}$	$1.31{\cdot}10^{3}$	$1.02{\cdot}10^{3}$	$1.85{\cdot}10^{3}$	2571	4659	2089
$1.07{\cdot}10^{-3}$	$7.61{\cdot}10^{-4}$	$1.38{\cdot}10^{-3}$	$9.35{\cdot}10^{2}$	$7.25{\cdot}10^{2}$	$1.31{\cdot}10^{3}$	3615	6553	2939
$1.50{\cdot}10^{-3}$	$1.07{\cdot}10^{-3}$	$1.94{\cdot}10^{-3}$	$6.65{\cdot}10^{2}$	$5.16{\cdot}10^{2}$	$9.35{\cdot}10^{2}$	5084	9212	4129
$2.11{\cdot}10^{-3}$	$1.50{\cdot}10^{-3}$	$2.73{\cdot}10^{-3}$	$4.73{\cdot}10^{2}$	$3.67{\cdot}10^{2}$	$6.65{\cdot}10^{2}$	7148	12954	5807
$2.97{\cdot}10^{-3}$	$2.11{\cdot}10^{-3}$	$3.83{\cdot}10^{-3}$	$3.36{\cdot}10^{2}$	$2.61{\cdot}10^{2}$	$4.73{\cdot}10^{2}$	10051	18217	8167
$4.18{\cdot}10^{-3}$	$2.97{\cdot}10^{-3}$	$5.39{\cdot}10^{-3}$	$2.39{\cdot}10^{2}$	$1.86{\cdot}10^{2}$	$3.36{\cdot}10^{2}$	14134	25614	11481
$5.88{\cdot}10^{-3}$	$4.18{\cdot}10^{-3}$	$7.58{\cdot}10^{-3}$	$1.70{\cdot}10^{2}$	$1.32{\cdot}10^{2}$	$2.39{\cdot}10^{2}$	19874	36020	16147
$8.27{\cdot}10^{-3}$	$5.88{\cdot}10^{-3}$	$1.07{\cdot}10^{-2}$	$1.21{\cdot}10^{2}$	$9.38{\cdot}10^{1}$	$1.70{\cdot}10^{2}$	27947	50649	22703
$1.16{\cdot}10^{-2}$	$8.27{\cdot}10^{-3}$	$1.50{\cdot}10^{-2}$	$8.60{\cdot}10^{1}$	$6.67{\cdot}10^{1}$	$1.21{\cdot}10^{2}$	39298	71222	31925
$1.64{\cdot}10^{-2}$	$1.16{\cdot}10^{-2}$	$2.11{\cdot}10^{-2}$	$6.12{\cdot}10^{1}$	$4.74{\cdot}10^{1}$	$8.60{\cdot}10^{1}$	55260	100152	44893
$2.30{\cdot}10^{-2}$	$1.64{\cdot}10^{-2}$	$2.96{\cdot}10^{-2}$	$4.35{\cdot}10^{1}$	$3.37{\cdot}10^{1}$	$6.12{\cdot}10^{1}$	77706	140834	63129
$3.23{\cdot}10^{-2}$	$2.30{\cdot}10^{-2}$	$4.17{\cdot}10^{-2}$	$3.09{\cdot}10^{1}$	$2.40{\cdot}10^{1}$	$4.35{\cdot}10^{1}$	109270	198038	88769
$4.55{\cdot}10^{-2}$	$3.23{\cdot}10^{-2}$	$5.86{\cdot}10^{-2}$	$2.20{\cdot}10^{1}$	$1.71{\cdot}10^{1}$	$3.09{\cdot}10^{1}$	153654	278480	124827
$6.39{\cdot}10^{-2}$	$4.55{\cdot}10^{-2}$	$8.24{\cdot}10^{-2}$	$1.56{\cdot}10^{1}$	$1.21{\cdot}10^{1}$	$2.20{\cdot}10^{1}$	216067	391599	175533
$8.99{\cdot}10^{-2}$	$6.39{\cdot}10^{-2}$	$1.16{\cdot}10^{-1}$	$1.11{\cdot}10^{1}$	$8.63{\cdot}10^{0}$	$1.56{\cdot}10^{1}$	303833	550663	246831
$1.26{\cdot}10^{-1}$	$8.99{\cdot}10^{-2}$	$1.63{\cdot}10^{-1}$	$7.91{\cdot}10^{0}$	$6.14{\cdot}10^{0}$	$1.11{\cdot}10^{1}$	427248	774340	347093
$1.78{\cdot}10^{-1}$	$1.26{\cdot}10^{-1}$	$2.29{\cdot}10^{-1}$	$5.62{\cdot}10^{0}$	$4.36{\cdot}10^{0}$	$7.91{\cdot}10^{0}$	600794	1088874	488081
$2.50{\cdot}10^{-1}$	$1.78{\cdot}10^{-1}$	$3.22{\cdot}10^{-1}$	$4.00{\cdot}10^{0}$	$3.10{\cdot}10^{0}$	$5.62{\cdot}10^{0}$	844834	1531168	686335

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